Next Steps for the Cut-and-Project Paper

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Purpose of this note

This document collects important hints, warnings, and concrete next steps for improving the manuscript

Period Length of One-Dimensional Cut-and-Project Sequences with Rational Slope and Arbitrary Real Window Width.

The central result appears to be a clean and potentially publishable formula for the period length of a rational one-dimensional cut-and-project construction. However, several issues should be clarified before submitting the paper to arXiv or to a journal.

Overall assessment

The work is potentially worth putting on arXiv, but not yet in exactly its current form.

- The mathematical value is real, but probably modest rather than major.
- The result is more naturally described as elementary combinatorics, discrete geometry, or symbolic dynamics than as deep number theory.
- The formula is attractive because it is exact, simple, and covers all rational slopes and all real window widths $\omega \geq 0$.
- The theorem is likely best framed as a clean explicit result about rational cut-and-project constructions, rather than as a major contribution to the general theory of aperiodic order.
- An arXiv preprint is reasonable after revision.
- A short journal note might be possible if the statement is correct, the proof is fully polished, and the relation to existing literature is handled carefully.

Central mathematical result

The main formula is:

$$\lambda = \begin{cases} N, & D \nmid N, \\ N/D, & D \mid N, \end{cases}$$

where

$$N = \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1, \qquad D = \alpha^2 + \beta^2.$$

This is a strong feature of the paper: the result is short, exact, and easy to state.

The proof reduces the geometry to arithmetic progressions modulo D. The key invariant is

$$r = \alpha x - \beta y,$$

and the projected coordinate is controlled by

$$\tilde{p} = \beta x + \alpha y.$$

For fixed r, the projected values lie in an arithmetic progression modulo D. The formula follows from counting how N consecutive values of r distribute among the residue classes modulo D.

Most important issue: multiset versus set

The manuscript currently defines the sorted projected values as a multiset. This is mathematically legitimate, but it must be stated extremely clearly.

In standard cut-and-project terminology, one usually speaks about point sets, where duplicate projected points are identified. In your manuscript, however, different lattice points may project to the same $\tilde{p}$-value, and these repeated values are counted with multiplicity.

This distinction is crucial.

Why this matters

In the degenerate case $D \mid N$, your proof says that every integer $\tilde{p}$ is achieved with multiplicity N/D. Hence the sorted sequence has the form

$$\dots, \underbrace{t, \dots, t}_{N/D}, \underbrace{t+1, \dots, t+1}_{N/D}, \dots$$

and the difference sequence has the form

$$\underbrace{0, \dots, 0}_{N/D-1}, 1, \underbrace{0, \dots, 0}_{N/D-1}, 1, \dots$$

This has period N/D.

But this is only true if multiplicities are retained. If duplicate projected points are collapsed, then the set of projected values is simply all integers, and the ordinary gap sequence has constant gap 1, hence period 1.

Therefore the theorem should explicitly be described as a theorem about the projected lattice-point multiset, or about the multiset gap sequence.

Recommended fix

Change the title, abstract, and definitions so that the word "multiset" appears prominently.

Possible titles:

- *Period Length of the Multiset Gap Sequence in Rational One-Dimensional Cut-and-Project Projections*
- *Period Lengths for Projected Lattice-Point Multisets from Rational Cut-and-Project Strips*
- *A Period Formula for Rational Cut-and-Project Strip Projections with Multiplicity*

Add a remark such as:

> The projected values are counted with multiplicity throughout this paper. Thus, if two distinct lattice points have the same projected coordinate, they contribute two consecutive entries to the sorted projected sequence and hence may produce a zero gap. If multiplicities are discarded, the resulting ordinary point-set gap sequence may have a smaller period. In particular, in the case $D \mid N$, the set-valued projected sequence has constant gap and period 1, whereas the multiset sequence has period N/D.

This clarification is essential before submission.

Tone and positioning

The paper should be presented as a clean explicit formula in a rational special case, not as a major breakthrough in aperiodic order.

The current introduction is acceptable in spirit, but it should be toned down slightly.

Recommended positioning:

> The irrational case is central in the theory of aperiodic order. The rational case is periodic and therefore often regarded as elementary or degenerate from that perspective. Nevertheless, the exact period length of the projected multiset gap sequence appears not to be commonly recorded in this explicit form. The present note gives a complete elementary formula.

Avoid statements that sound too grand, such as suggesting that the paper fills a major gap in the theory of model sets. The result is valuable because it is exact and explicit, not because it revolutionises the field.

arXiv suitability

The paper is suitable for arXiv after revision.

Recommended arXiv categories:

- Primary: `math.CO` or `math.DS`
- Possible cross-listing: `math.NT`, `math.MG`

The work is probably not best submitted primarily as number theory. It uses number-theoretic tools such as congruences, residue classes, and coprimality, but the main object is a gap sequence arising from a geometric/combinatorial construction.

Important arXiv warning: AI co-authors

Do not list AI systems as co-authors.

The current manuscript lists Gemini and Claude as authors. This should be changed before submission.

Use:

```
\author{Dirk Kunert}
```

Then move the AI discussion to the acknowledgements.

Suggested wording:

> The author used ChatGPT, Gemini, and Claude as AI-assisted tools for exploration, drafting, proof checking, and Lean formalisation. The author has independently checked the mathematical content and accepts full responsibility for the results and any errors.

Avoid wording such as "Gemini derived and proved the formula" in the author block. That makes the paper look less professionally controlled. The responsible author should be the human mathematician.

Suggested revised abstract

A safer and more professional abstract would be:

> We study the gap sequence of the projected lattice-point multiset obtained from a rational-slope strip in the plane. Let the slope be α/β, with $\gcd(\alpha, \beta) = 1$, and let the window parameter be $\omega \geq 0$. Counting projected points with multiplicity, the sequence of consecutive gaps is periodic. We prove that its minimal period is
>
> $$\lambda = \begin{cases} N, & D \nmid N, \\ N/D, & D \mid N, \end{cases}$$
>
> where
>
> $$N = \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1, \qquad D = \alpha^2 + \beta^2.$$
>
> The proof reduces the geometry to the distribution of N consecutive residue classes modulo D. In the non-uniform case, the heavy residue classes form a cyclic interval with trivial stabiliser, which implies minimality of the period. We also indicate how the formula changes when multiplicities are discarded.

Proof issues to check carefully

The proof is plausible, but the following points should be checked line by line.

1. Sorting of a bi-infinite multiset

The manuscript sorts the projected values into a bi-infinite sequence

$$(\tilde{p}_s^{(i)})_{i\in\mathbb{Z}}.$$

This should be justified carefully. One needs to know that the projected values are locally finite as a multiset and can be indexed increasingly over $\mathbb{Z}$.

Suggested addition:

> Since the projected values are a finite union of arithmetic progressions of common difference D, counted with finite multiplicities, the resulting multiset is locally finite and unbounded in both directions. Hence it admits a nondecreasing enumeration indexed by $\mathbb{Z}$, unique up to translation of the index.

2. Multiplicity convention

State whether the sorted sequence is strictly increasing or nondecreasing. With multiplicities, it is nondecreasing.

Use "nondecreasing" rather than "ascending" if duplicates are present.

3. The assertion $\lambda \mid N$

The proof says that the sequence has a repeating block of length N, hence $\lambda \mid N$. This is correct if the block structure is precisely periodic. It should be made explicit that after N entries the sorted multiset is translated by D:

$$\tilde{p}_s^{(i+N)} = \tilde{p}_s^{(i)} + D.$$

Then

$$\delta^{(i+N)} = \delta^{(i)}.$$

This gives N as a period, hence the minimal period divides N.

4. The minimality argument

The minimality proof is the heart of the paper. It is good, but it should be made maximally rigorous.

The argument is:

1. Suppose $\lambda' < N$ is a period.
2. Since N is a period, the minimal period divides N, so $N = k\lambda'$ for some $k > 1$.
3. Let σ be the sum of one λ'-block of gaps.
4. Then translation by σ preserves the multiset.
5. Since N gaps sum to D, we get $k\sigma = D$.
6. Therefore translation by σ induces a nontrivial stabiliser of the residue multiplicity function modulo D.
7. But the heavy set is a proper cyclic interval, and such an interval has trivial stabiliser.
8. Contradiction.

This is elegant. The paper should isolate this as a lemma, perhaps:

Lemma 1. *Let S be a finite union of arithmetic progressions of common difference D, with residue multiplicity function μ. If the gap sequence has period L, then the translation by the sum of one L-block of gaps preserves μ.*

This would make the proof more modular.

5. The cyclic interval stabiliser

The statement that a proper nonempty cyclic interval has trivial stabiliser is correct, but the proof should be made more formal.

Suggested lemma:

Lemma 2. *Let $D \geq 2$, and let*

$$H = \{a, a+1, \ldots, a+s-1\} \subset \mathbb{Z}/D\mathbb{Z}$$

with $0 < s < D$. If $H + \tau = H$, then $\tau \equiv 0 \pmod D$.

A clean proof uses the boundary of H: the set H has exactly two boundary edges in the cyclic graph C_D, whereas a nontrivial translation moves the boundary. Alternatively, use cardinality of intersections.

Lean formalisation

The Lean formalisation is a strength, but the wording should be precise.

Do not overclaim that the entire geometric theorem has been formalised from first principles if the construction is partly abstracted through a typeclass.

Recommended wording:

> The residue-class and minimal-period argument has been formalised in Lean 4. The geometric reduction from the strip construction to the residue multiset is represented at the combinatorial level. Thus the formalisation verifies the algebraic core of the proof rather than the full analytic geometry from first principles.

If the repository is public, make sure that:

- it builds from a clean clone;
- the Lean version is specified;
- the Mathlib commit is specified;
- there are no `sorry`;
- instructions are included in a `README.md`;
- the file names and line numbers in the paper are still correct.

Computational verification

The computational verification is useful, but it should not be overemphasised. Once the proof is complete, the tests are supporting evidence, not part of the mathematical proof.

Recommended placement:

- keep a short remark in the main text;
- move details to an appendix or repository;
- describe the testing method clearly enough to be reproducible.

Mention:

- ranges of α, β, ω;
- how ω was sampled;
- how the period was computed;
- whether multiplicities were counted;
- how degenerate cases $D \mid N$ were generated.

Literature section

The related-literature section should be strengthened.

The current discussion of the three-distance theorem, Christoffel words, and Sturmian sequences is useful, but the paper should avoid relying on Wikipedia as a research citation.

Recommended changes:

- Replace the Wikipedia citation for the three-distance theorem with a mathematical source.

- Keep Berstel–Lauve–Reutenauer–Saliola for Christoffel words.
- Keep Haynes–Koivusalo–Walton–Sadun for cut-and-project gap problems.
- Add a sentence explaining exactly why these papers do not contain your formula.
- Be cautious with the claim "no prior work was found". Better: "The author is not aware of a reference that records this exact period formula in the above multiset convention."

Suggested wording:

> The author is not aware of a reference that states the above formula for the minimal period of the multiset gap sequence associated with a rational one-dimensional cut-and-project strip. The result is elementary once the appropriate residue-class description is introduced, but it appears useful to record explicitly.

This is modest and credible.

Possible mathematical extensions

The current open problems are good. They can be sharpened.

1. Window translation

Study what happens if the window is shifted transversally. Does the same formula depend only on the number N of admissible invariant values, or do endpoint effects change the period?

2. Set-valued version

Derive the exact period when projected points are counted without multiplicity. This may be a natural companion theorem and would remove possible confusion with standard model-set terminology.

3. Gap-value distribution

The paper determines the period length, but not the actual distribution of gap sizes within one period.

A useful extension would be:

> For fixed α, β, ω, determine the distinct gap values and their multiplicities in one minimal period.

This would connect more directly to the three-distance theorem and gap problems.

4. Higher dimensions

This is a natural but much harder direction. It should be described as speculative.

Recommended structure for a revised paper

A cleaner revised structure could be:

1. Introduction
2. Statement of the multiset convention

3. Geometry-to-residue reduction
4. The residue multiplicity function
5. Generic case $D \nmid N$
6. Uniform case $D \mid N$
7. Set-valued comparison
8. Computational checks
9. Lean formalisation
10. Related literature and open problems

Concrete checklist before arXiv

Before submitting to arXiv, do the following:

1. Remove AI systems from the author list.
2. Put AI assistance only in the acknowledgements.
3. Add an explicit multiset convention near the beginning.
4. Change the title to mention "multiset" or "with multiplicity".
5. Replace "ascending" by "nondecreasing" where multiplicities are possible.
6. Add a remark explaining the difference between the multiset gap sequence and the ordinary set-valued gap sequence.
7. Make the proof of $\tilde{p}_s^{(i+N)} = \tilde{p}_s^{(i)} + D$ explicit.
8. Isolate the minimality argument as a clean lemma.
9. State and prove the trivial-stabiliser lemma formally.
10. Tone down claims of novelty.
11. Replace Wikipedia citations with mathematical references.
12. Check all bibliography entries.
13. Check all Lean repository links.
14. Ensure the Lean code builds from a clean clone.
15. Make clear exactly what the Lean proof does and does not formalise.
16. Consider adding a short section on the version without multiplicities.
17. Decide the arXiv category, probably `math.CO` or `math.DS`, with possible cross-listing to `math.NT`.
18. Compile the paper locally and check for overfull boxes, undefined references, and bibliography problems.
19. Ask at least one human mathematician to read the proof.

Final judgement

The work is worth developing.

It is not likely to be regarded as a major new theorem in number theory, but it can be a respectable and useful short mathematical note. Its strengths are:

- a clean exact formula;
- a transparent residue-class proof;
- coverage of all real $\omega \geq 0$;
- computational verification;
- partial Lean formalisation.

Its weaknesses are:

- possible confusion between point sets and multisets;
- elementary depth;
- need for better literature positioning;
- problematic AI co-author presentation;
- risk of overclaiming the scope of the Lean formalisation.

With the revisions listed above, the paper is suitable for arXiv as a modest, clear, and potentially useful contribution.